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EYS-165

B. Tech. (New Scheme) Examination, 2024

(Fourth Semester)

SIGNAL AND SYSTEMS

EL-404

Time : 3 Hours]

[Maximum Marks : 60

Note : Attempt all questions. All questions carry equal marks.

1. (a) What is meant by System ? Classify the different types of System.
(b) Explain exponential and sinusoidal signals.

Or

- (a) Explain static and dynamic systems.
(b) Explain deterministic and random signals.
2. (a) What is meant by Fourier Series ?
(b) What is Discrete time Fourier transform ? Explain its properties.

Or

- (a) Explain duality sinc and signum function.
 - (b) Compare between Sampling theorem and Aliasing.
3. (a) Explain the principle of Laplace transform.
- (b) Explain the properties of inverse laplace transform.

Or

- (a) Explain about to state variable analysis.
 - (b) Explain ROC in laplace transform.
4. (a) What is the basic formula for the Z-transform ?
- (b) Write concept phase delay.

Or

- (a) Explain ROC in Z-transform.
 - (b) Explain the properties of inverse Z-transform.
5. (a) What is meant by Random Process ? Explain.
- (b) Explain about cross-correlation function.

Or

- (a) Explain the different types of Random Variable.
- (b) Explain about to cumulative distribution function and probability density function.

6. Explain the following (any *three*) :

- (a) Periodic and aperiodic signals
- (b) Continuous time Fourier transform.
- (c) Solution of state equation
- (d) Group delay
- (e) Ergodicity.